

Column Design

in ProtaStructure

Placement | Reinforcement | Analysis | Design Check | Schedule

What You Will Learn

- Column types: rectangular, square, circular, and L-shaped — when to use each
- Placing columns on structural grids and setting section dimensions in ProtaStructure
- Reinforcement rules: minimum/maximum steel ratios, cover, bar spacing, stirrup design
- Running column design, reading Unity Check results, and generating the column schedule

Column Slenderness Check (EC2)

$$\lambda = L_{\text{eff}} / i \leq \lambda_{\text{lim}} \text{ (EC2 Cl. 5.8.3)}$$

L_{eff} = effective length; i = radius of gyration = $\sqrt{I/A}$; $\lambda_{\text{lim}} \approx 26\text{--}120$ depending on loads

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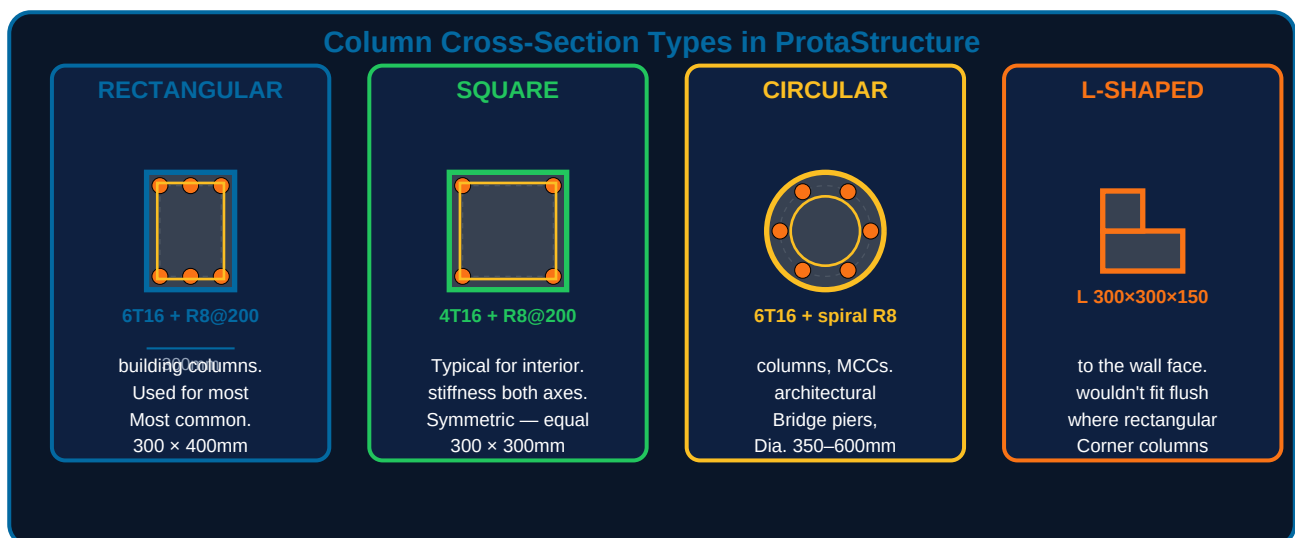
PROTA-006: Column Design in ProtaStructure

Columns are the vertical load-carrying members that transfer gravity loads from floors and beams down to foundations. In a reinforced concrete frame, columns also resist lateral forces (wind and seismic) through their stiffness and moment capacity. Correctly sizing and reinforcing columns is critical — an undersized column can cause progressive collapse of the entire structure. ProtaStructure automates the design calculation while the engineer controls section selection, material grades, and covers all code checks.

Learning Objectives

- Identify the four main column cross-section types used in ProtaStructure
- Place columns on a structural grid and set dimensions, concrete grade, and reinforcement cover
- Understand how ProtaStructure analyses columns under axial force (N), bending moment (M), and shear (V)
- Run column design, interpret Unity Check results, and modify sections that fail the check
- Generate a column schedule and export reinforcement drawings from ProtaStructure

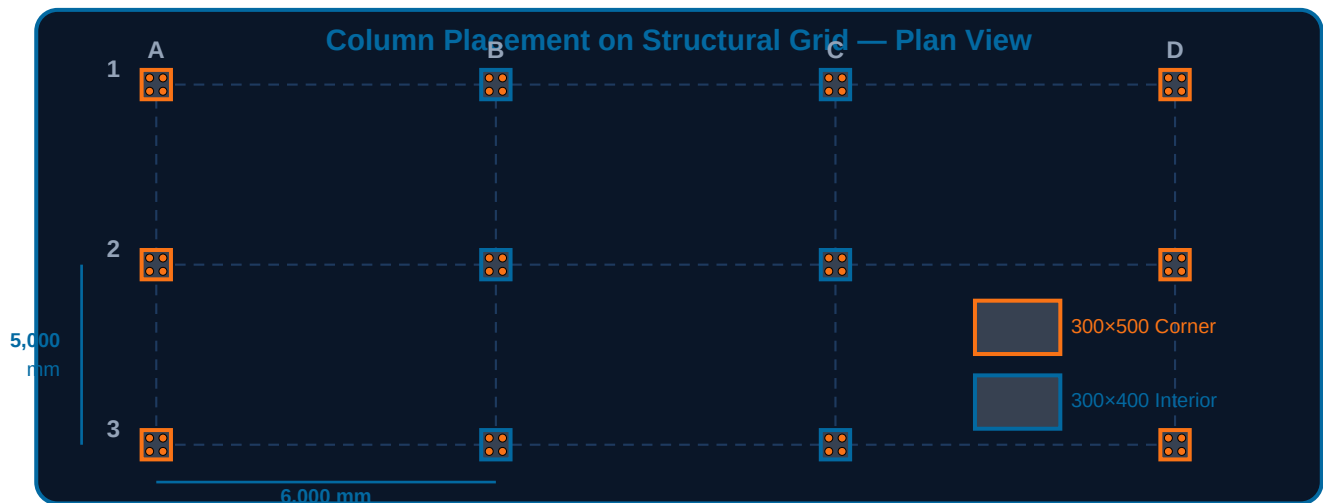
1. Column Types in ProtaStructure



Column Type	Common Dimensions	Typical Use	ProtaStructure Input
Rectangular (b × h)	250×300 to 400×600 mm	Most building columns. Size both sides for biaxial bending resistance.	Width (b) and Depth (h) in Properties. Orientation set with rotation angle.

Square (b = h)	250×250 to 500×500 mm	Interior columns with equal spans in both directions. No preferred bending axis.	Set b = h in Properties. ProtaStructure treats as rectangular with equal sides.
Circular (diameter)	Ø300 to Ø600 mm	Exposed architectural columns, bridge piers, car park columns.	Select "Circular" column type → input diameter. Spiral or link reinforcement option.
L-shaped / Irregular	Custom	Corner columns at re-entrant corners, columns in confined plan locations.	Use "Custom Section" and draw the polygon shape using coordinate input.

2. Placing Columns on the Structural Grid



How to Place Columns in ProtaStructure

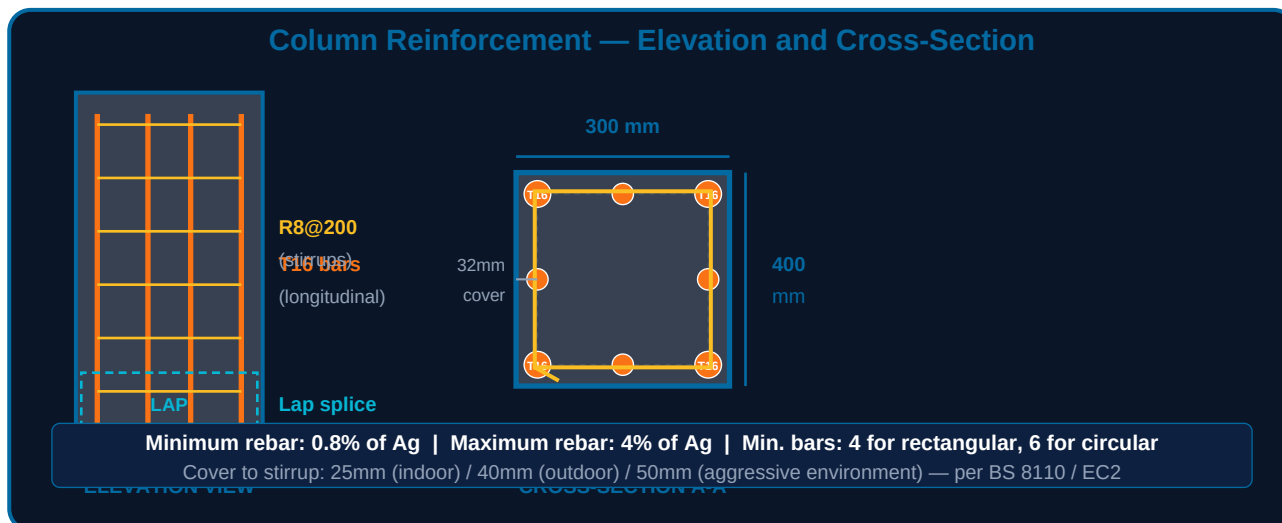
Step	Action in ProtaStructure	Notes
1	Open floor plan view for the relevant storey	Columns are placed storey by storey. Select the correct storey from the Storeys panel.
2	Home tab → Column (keyboard: C)	The Column cursor activates. The Properties panel on the right shows column settings.
3	Set column properties before placing: Width, Depth, Concrete Grade, Cover	Set these in Properties panel. They apply to all columns placed in this session until changed.
4	Click on grid intersections to place columns	Snap to grid points is automatic when grid is displayed. Columns appear as filled rectangles.
5	Set column orientation (rotation)	After placing, select column → Properties → Rotation angle (degrees). Or use the Rotate tool.
6	Verify placement in 3D view	3D tab → Perspective. Columns should align through all storeys and connect to beams and slabs.

Column Properties Reference

Property	Typical Value	Notes / Code Requirement
Width (b)	250 – 600 mm	Minimum 200mm for structural columns per most codes. Prefer multiples of 50mm.

Depth (h)	250 – 600 mm	Usually $h \geq b$. For corner columns, larger h in the longer span direction.
Concrete Grade	C25/30 to C40/50	C30/37 most common for RC columns in Ethiopia and East Africa. Specify fck (cylinder) in EC2.
Rebar Grade	B500B (EC2) / T500 (BS)	$f_y = 500$ MPa for high-yield deformed bars. ProtaStructure sets yield strength automatically from bar type.
Cover (main bars)	25 mm indoor, 40 mm outdoor	EC2 Table 4.4N. Minimum cover to main longitudinal bars (not to stirrups/links).
Cover (to stirrup)	15–25 mm less than main bar cover	Stirrup cover = main bar cover minus the stirrup bar diameter. ProtaStructure calculates this automatically.
Effective length factor k	0.5 (both ends fixed) to 2.0 (cantilever)	$k = 0.7$ for columns braced against sway with one end pinned (typical floor-to-floor). ProtaStructure computes from boundary conditions.

3. Column Reinforcement Design



Reinforcement Rules for RC Columns (EC2 / BS 8110)

Reinforcement Parameter	Minimum	Maximum	Reason
Longitudinal steel ratio (ρ)	$0.002 \times A_c$ (0.2%)	$0.04 \times A_c$ (4%)	Min: prevent concrete cracking from shrinkage. Max: avoid concrete placement difficulty and bar congestion.
Minimum number of bars	4 bars for rectangular, 6 bars for circular	No maximum, but prefer even numbers	Ensures symmetrical reinforcement for biaxial bending resistance.
Minimum bar diameter	12 mm	Typically 32 mm	Smaller bars are impractical for column main steel — they do not provide adequate stiffness.
Maximum bar spacing	300 mm between bars on one face	—	EC2 Cl. 9.5.2. Prevents concrete from failing in between bars under compression.
Stirrup / link spacing	min: $20 \times \phi_{\text{min}}$ (longitudinal bar)	Max: $0.6 \times h$ (column height) or 400 mm	Stirrups prevent buckling of longitudinal bars under compression. More stirrups in lap splice zones.
Lap splice length	$\geq 1.4 \times l_{bd}$ (design anchorage length)	—	Increased to $1.4\times$ at lap splices. Typically 500–800mm for T16 bars in C30 concrete.

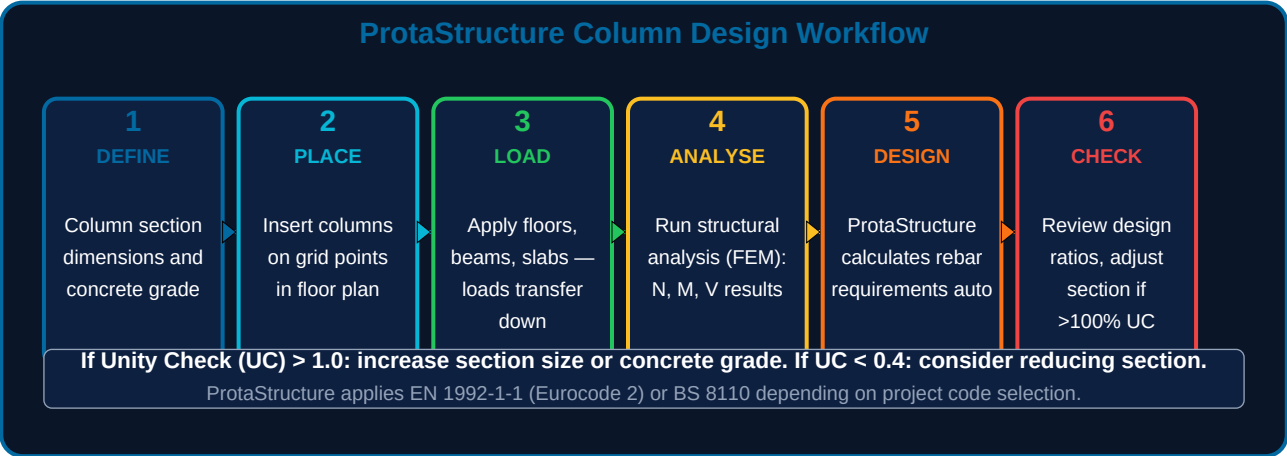
Selecting Reinforcement in ProtaStructure

ProtaStructure can design reinforcement automatically (it calculates the required area of steel and selects bar sizes and numbers from the bar schedule you define), or you can specify bars manually and check whether they satisfy the code requirements.

Design Mode	How to Select	Output
Auto Design	Design tab → Design Columns → Auto Design All	ProtaStructure selects number and size of bars to satisfy N-M interaction and minimum requirements. Review result in column schedule.
Manual Design	Select column → Properties → Longitudinal Bars: specify "8T20" → Run Check	ProtaStructure checks the manually specified bars against all code requirements and shows Unity Check ratios.
Design Parameters	Design tab → Column Design Parameters	Set preferred bar sizes, minimum/maximum bars, link spacing rules. These preferences guide Auto Design.

4. Column Analysis — What ProtaStructure Calculates

Before designing reinforcement, ProtaStructure performs a linear elastic finite element analysis of the entire structural model. Every column is analysed for the most critical combination of forces from all load cases.



Column Design Forces

Force / Moment	Symbol	What It Represents	Design Impact
Axial compressive force	N_Ed (kN)	Total vertical load on the column from all floors above, including self-weight	Dominant force in lower storey columns. Increases required steel area and concrete grade.
Bending moment — major axis	M_Ed,y (kN·m)	Bending from beams framing in at the column, wind/seismic lateral forces, and eccentric loading	Combined with N_Ed in N-M interaction diagram check. Requires bars on the tension face.
Bending moment — minor axis	M_Ed,z (kN·m)	Same as M_Ed,y but about the perpendicular axis. Most columns are biaxially bent.	EC2 requires biaxial bending check. Corner columns often have significant M in both axes.
Shear force	V_Ed (kN)	Horizontal force from lateral loads and frame action	Governs stirrup spacing in the critical region (usually top and bottom ¼ of column height).

Reading Column Design Results in ProtaStructure

Result / Display	What It Shows	Action Required
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Unity Check (UC) ratio	Ratio of applied forces to capacity. UC = 1.0 means exactly at limit.	UC < 1.0: PASS (green). UC ≥ 1.0: FAIL (red) — increase section or grade.
N-M Interaction diagram	Graph showing the capacity envelope of the column cross-section with the design point plotted inside or outside.	Design point must be inside the envelope. If outside: fail.
Reinforcement details	Number, size, and arrangement of bars selected by the program.	Check for practicality: will bars fit with required cover and spacing?
Slenderness check	Column slenderness ratio $\lambda = L_{eff}/i$ compared to λ_{lim} .	If $\lambda > \lambda_{lim}$: column must be designed for second-order effects. ProtaStructure handles this automatically.
Column schedule table	Summary of all columns: storey, reference, section, bars, links.	Export to PDF or DWG for structural drawings.

5. Practice Exercises

#	Exercise	Instructions
1	Place columns on a 3-bay grid	- Set up a 3-bay × 3-bay grid: bays of 6,000mm × 5,000mm. - Place 300×400mm columns at all 16 intersections (4×4 grid). - Change corner columns to 300×500mm — larger depth parallel to the shorter span.
2	Minimum reinforcement check	Column: 300×400mm = $A_c = 120,000 \text{ mm}^2$. Concrete C30/37, Steel B500B. Calculate: (a) min steel area = $0.2\% \times A_c$, (b) max steel area = $4\% \times A_c$. How many T16 bars (area 201mm ² each) are needed for minimum? How many for maximum?
3	Slenderness calculation	Column: 300×400mm, storey height 3,500mm, pinned-fixed ($k=0.7$). Calculate: (a) $L_{eff} = k \times L$, (b) $I = bh^3/12$ (minor axis), (c) $A = b \times h$, (d) radius of gyration $i = \sqrt{I/A}$, (e) $\lambda = L_{eff}/i$. Is $\lambda < 26$ (short column limit for braced frames with equal moments)?
4	Auto design and check	- In ProtaStructure: run the full structural analysis on your model. - Design tab → Design Columns → Auto Design All. - View results in 3D. Identify any red (failed) columns and note their location. - For each failed column, increase the section by 50mm in both directions and re-run.
5	Column schedule review	- After successful design, Generate → Column Schedule. - Verify that: all columns show $UC < 1.0$; bar spacing $\leq 300\text{mm}$; link spacing $\leq 400\text{mm}$. - Export the column schedule as a PDF and check the reinforcement summary matches your expectations.

Lesson Summary

Key Concept	Summary Rule
Column types	Rectangular: most common. Square: interior with equal spans. Circular: architectural. L-shaped: corner constraints. ProtaStructure supports all via Properties.
Minimum dimensions	250mm minimum in any direction for structural RC columns. Prefer multiples of 50mm to match formwork.
Concrete grade	C30/37 ($f_{ck}=30\text{MPa}$) typical for residential/commercial. Use C35/45 or above for heavily loaded lower storey columns.
Steel ratio	0.2% minimum (shrinkage cracking). 4% maximum (congestion). Aim for 1–2.5% in most practical designs.
Cover	25mm indoor, 40mm outdoor (to main bars, per EC2). ProtaStructure uses these as default — verify in project settings.
Unity Check	$UC < 1.0$ = PASS. $UC \geq 1.0$ = FAIL. If $UC > 1.2$: increase section by 50mm each way. If $UC < 0.5$: section may be over-designed — consider reducing.

Slenderness	Short column ($\lambda < \lambda_{lim}$): no second-order effects. Slender ($\lambda > \lambda_{lim}$): ProtaStructure amplifies moments automatically per EC2 Cl. 5.8.
Column schedule	Generate after all columns pass design check. Column schedule is a formal deliverable in structural engineering packages.

What Is Next — PROTA-007

PROTA-007: Slab Design — One-way and two-way flat slabs and ribbed slabs in ProtaStructure. Slab panel creation, thickness selection, and load application. Punching shear check at column heads. Slab reinforcement design — top steel over supports, bottom steel at midspan. Generating slab drawings and reinforcement schedules.

Resource	Details
PROTA PDF Series	PROTA-001 through PROTA-030 — ayetechub.com/pdfs.html
ProtaStructure Trial	Free 30-day trial — protasoftware.com/download
Community	Telegram: t.me/ayetechub